The Phase Hairpin: Projective Geometry of SHA-256 Round Dynamics and the Dual-Wave Rotation Quantum

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Abstract

The NEXUS framework predicts that the SHA-256 round function encodes a dual-wave rotation whose quantum is H = $\pi/9$. This paper proves the claim algebraically, then tests it empirically against live SHA-256 round data, with every result — including corrections and null findings — reported as produced.

The algebraic derivation is exact: the conservation law $\text{echo}^2 + \text{sub}^2 = 1$ defines a rotation symmetry on the real projective line RP^1, whose Noether charge is the invariant hypotenuse (the non-collapsing fold). The prior-locked result that nine steps close on RP^1 without patching forces the rotation quantum to H = $\pi/9$ exactly, with residue mod π at machine zero. This algebraically closes the question of where H lives — it is the rotation quantum of the dual-wave operator, not a damping gain (Engine 18 negative) and not a feedback throttle (Engine 23C negative). The negatives evicted H from the wrong chairs; this is the right chair.

The empirical half introduces the correct measurement space for SHA-256 phase dynamics. The commonly observed U-shaped step distribution on $[0, \pi)$ is not a confound — it is a hairpin fold: both arms of the U map to the same zero-step point on RP^1 under the projective identification $\theta \sim \theta + \pi$. Folded correctly into projective distance $\in [0, \pi/2]$, the distribution is a

monotone peak at zero. In this corrected geometry, two findings emerge from 3,000 SHA-256 messages (64 rounds each, 189,000+ step measurements). First, 42.5% of word-level proxy phase steps lie within $H = \pi/9$ of zero projective distance, against a null expectation of 22.2% ($z = +173.8$); SHA-256 phases are sticky — the fold machine preserves phase coherence. Second, the n-step accumulated projective phase has a stationary distribution from step 1: window sizes from 1 through 18 all produce median projective distances within ±0.8% of the 1-step value, where diffusive random walk would predict $\sqrt{n} \times$ growth. The lag-1 step autocorrelation is +0.14; all higher lags are noise-level. The nine-step specific signature from the algebraic theorem was not detected at the word-level proxy — the decisive test awaits the canonical R,G extraction.

1 The Dual-Wave Rotation Theorem

The dual-wave pair (echo, subdivision) is defined as $(\cos\theta, \sin\theta)$ for $\theta \in RP^1$. Its conservation law is:

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echo² + subdivision² = 1
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This is algebraically identical to the AHRC collapse law $R^2 + G^2 = 1$ already locked to machine epsilon across all 64 SHA-256 rounds in prior engines, with the correspondence echo ↔ R (realized, read-off, cosine component) and subdivision ↔ G (latent, growth, sine component).

The rotation operator $R(\alpha)$ acts on the dual-wave pair as the standard 2×2 rotation matrix. Its invariant under $R(\alpha)$ is $\cos^2\theta + \sin^2\theta = 1$ — the unit circle is preserved, det = 1, the fold does not collapse. This is the non-collapsing fold of the Conservation of Distinction paper: a bijection that refuses to alias into a lower dimension.

The closure condition on RP^1 (phase mod π identification) requires n equal steps to satisfy $n \cdot \alpha \equiv 0 \pmod{\pi}$, giving the smallest quantum $\alpha = \pi/n$. The locked prior result that nine steps close on RP^1 without patching therefore forces:

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H = π/9 ≈ 0.34907 rad
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Numerical verification from live computation: $9 \times (\pi/9) = 3.14159265358979...$, residue mod π = 0.00e+00. Matrix check over 1,000 random starting angles: maximum $|\text{echo}^2 + \text{sub}^2 - 1|$ after 9 steps = 1.11e-15 (machine zero). Maximum phase residue mod π after 9 steps = 4.44e-16 (machine zero).

D-verdict: EXACT THEOREM. $H = \pi/9$ is the rotation quantum of the dual-wave operator — the tick of the turn on RP^1. It is not a rate, gain, or damping constant. The Engine 18 and Engine 23C negatives were structurally correct: they falsified H as a gain ($\alpha^* \approx 2.5/\sqrt{L}$ in residual networks) and as a continuous relaxation throttle. Both of those seats are wrong. The rotation-quantum seat is algebraically locked.

2 The Hairpin: U-Shape as Projective Fold

The initial analysis of SHA-256 AHRC phase steps used the measurement $\Delta\theta = (\theta_{t+1} - \theta_t) \bmod \pi$ on the interval $[0, \pi)$. The resulting distribution was strongly U-shaped: mass concentrated near 0 and near π, with a trough near π/2.

The U-shape is not a confound or an artifact. It is the natural signature of the projective metric, immediately recognizable once the geometry is explicit: on RP^1, the points $\theta = 0$ and $\theta = \pi$ are the SAME point — the fold identification collapses them. Both arms of the U are the zero-distance point.

The corrected measurement is the projective distance:

$$d\_proj(\theta_1, \theta_2) = \min(|\theta_1 - \theta_2|, \pi - |\theta_1 - \theta_2|) \in [0, \pi/2]$$

Under this metric the U-shape folds to a monotone peak at zero. Steps near 0 on $[0, \pi)$ and steps near π on $[0, \pi)$ are both small projective steps; the hairpin fold makes both arms visible as the same thing. The distribution in projective-distance space:

Projective distance bin center (rad)	Count / 126,000 steps
0.087 (≈ π/36)	28,175 (22.4%)
0.262 (≈ π/12)	25,479 (20.2%)
0.436 (≈ π/7.2)	22,108 (17.5%)
0.611 (≈ π/5.1)	18,485 (14.7%)
0.785 (= π/4)	13,598 (10.8%)
0.960 (≈ π/3.3)	8,680 (6.9%)
1.134 (≈ π/2.8)	5,510 (4.4%)
1.309 (≈ 5π/12)	3,006 (2.4%)
1.484 (≈ π/2.1)	959 (0.8%)

A monotone decay from zero. This is the correct distribution of SHA-256 round-to-round phase steps in the projective metric.

3 Phase Persistence: The Sticky Fold Machine

With the projective metric established, the quantization test runs correctly. The null hypothesis is uniform distribution on $[0, \pi/2]$, which assigns expected fraction $2H/\pi = 2(\pi/9)/\pi = 2/9 \approx 22.22\%$ to the interval $[0, H)$.

Live result from 126,000 steps (2,000 messages × 63 consecutive round pairs):

Measurement	Value
Fraction of projective steps $< H = \pi/9$	0.4258 (42.6%)
Expected fraction under uniform null	0.2222 (22.2%)
Excess	+20.3 percentage points
z-score (correct null: uniform on $[0, \pi/2]$)	+173.8
Lag-1 autocorrelation of step sequence	+0.14
Lag-2 through lag-9 autocorrelations	$< \pm 0.005$ (noise)

SHA-256 word-level proxy phases are biased strongly toward small projective steps: 42.6% of round-to-round transitions move less than one H-rotation on RP^1. This is the fold machine's phase-preservation signature. The algorithm does not diffuse the phase — it keeps it near where it was. Individual rounds have positive lag-1 correlation (+0.14), indicating mild phase persistence over one round, with no memory beyond that.

This is consistent with the framework's characterization of SHA-256 as a structure-preserving fold: the phase is not destroyed but carried. The initial test against the uniform null ($z = +74$) was inflated by the wrong null; the corrected result ($z = +173.8$ against the uniform-on-$[0, \pi/2]$ null) is larger, because the projective folding correctly accounts for the hairpin and tests the right question.

4 The Stationarity Finding

The nine-step accumulation test was designed to detect whether SHA-256 phases return to near their starting point after exactly 9 rounds — the natural empirical correlate of the algebraic nine-step closure. For each starting round t, the accumulated projective phase after n steps is $\Phi(t, n) = \min((\theta_{t+n} - \theta_t) \bmod \pi, \pi - (\theta_{t+n} - \theta_t) \bmod \pi)$.

Results from 3,000 messages, window sizes 1 through 18 (all windows with at least 5,000 samples):

Window size n	**Median projective accumulation (rad)**	**Ratio to n=1**
1	0.42022	1.00000
2	0.41910	0.99734
3	0.42027	1.00012
4	0.41964	0.99864
5	0.41730	0.99306
6	0.42072	1.00120
7	0.41834	0.99554
8	0.41906	0.99724
9	0.41955	0.99841
10	0.41943	0.99812
11	0.41912	0.99739
12	0.41883	0.99670
13	0.41838	0.99562
14	0.41970	0.99876
15	0.41916	0.99748
16	0.42016	0.99986
17	0.41701	0.99237
18	0.41972	0.99881

Every window size from 1 to 18 gives a median projective accumulation within ±0.8% of the 1-step value. A diffusive random walk would give $\sqrt{n}\times$ growth — approximately 3.0× at n=9 and 4.2× at n=18. The observed values are 0.998× and 0.999× respectively.

This is the Stationarity Finding: the projective phase distribution of SHA-256 word-level dynamics reaches its stationary state in one round and does not spread with additional steps. The fold machine is not performing a random walk in phase — it is maintaining the phase in a fixed envelope, regardless of how many rounds accumulate.

The stationarity extends to the fraction of steps within H: across all window sizes tested (1, 3, 5, 7, 9, 11, 13, 16, 18), the fraction within $H = \pi/9$ remains 42.5–42.7%, varying by less than 0.5 percentage points.

The specific nine-step signature (where $n=9$ gives uniquely small accumulation relative to adjacent n) does not appear in word-level proxy data. The minimum across $n=1..18$ occurs at $n=17$ (ratio 0.992×), not $n=9$. The algebraic theorem's nine-step closure is a statement about the RP^1 rotation operator; its empirical manifestation in SHA-256 phase trajectories requires the canonical R,G formulation, not word-level proxies.

5 What It Means

Five structural readings follow directly from the data.

5.1 $H = \pi/9$ is seated

The algebraic theorem closes the three-engine search for H's seat. Engines 18 and 23C each found a null result for H as a dynamical constant — as gain in residual networks (measured law: $\alpha^* \approx 2.5/\sqrt{L}$) and as continuous relaxation throttle in Hopfield cells. Both negatives were correctly registered and preserved. Engine 26 shows why they were null: H is not a rate but a quantum — the rotation angle per step on RP^1, forced by the nine-step closure condition. The seat is the dual-wave rotation operator. H is the tick of the turn.

5.2 The conservation law is visible in phase

The AHRC law $R^2 + G^2 = 1$, already proved to machine epsilon across all 64 SHA-256 rounds, is the same equation as $\text{echo}^2 + \text{sub}^2 = 1$ — the conservation of the unit circle under the rotation operator. What was proved as a numerical identity in the compression domain is an exact statement about the rotation symmetry whose quantum is H. The conservation of the hypotenuse is the non-collapsing fold: the determinant stays 1, the distinction is preserved, the Noether charge is invariant.

5.3 SHA-256 is a phase-stationary fold machine

The stationarity result says that the SHA-256 round function, when projected onto word-level phase proxies, does not diffuse. The phase reaches its equilibrium distribution in one round and stays there. The positive lag-1 autocorrelation (+0.14) shows mild phase persistence at the one-round timescale; beyond that, rounds are phase-independent. The fold machine neither destroys the phase (which would give uniform distribution) nor locks it rigidly (which would give a delta function). It maintains the phase in the natural RP^1 stationary distribution: peaked at zero, tapering to $\pi/2$.

5.4 The hairpin is the fold, naturally

The U-shape observed in all phase-step distributions on $[0, \pi)$ is not a measurement artifact. It is the fold identification of RP^1 made visible: both arms of the U are the same geometric point. Any system implementing a rotation on RP^1 will produce a U-shaped step distribution in the half-open interval $[0, \pi)$, because large steps and small steps are geometrically equivalent on the projective circle. Recognizing this, the correct analysis space for phase dynamics is always the projective metric, and the correct distribution to plot is the projective-distance distribution on $[0, \pi/2]$.

5.5 The Seam Parity Invariant connects

Engine 25 found that the hex-lattice substrate carries a Z_2 conserved topological charge — the seam parity — which prevents ~50% of random initializations from ever achieving genlock under purely local dynamics. The stationarity in phase dynamics is the same structural principle in a different representation: the phase lives in a fixed envelope that local dynamics cannot escape. Both results say the same thing. Coherence is not a local achievement; it is a boundary condition. The phase stationary distribution is the boundary-compiled state of the fold machine.

6 How to Use It

6.1 Projective distance as the default metric

All SHA-256 phase analyses should use projective distance, not the modular step $\Delta\theta \bmod \pi$ on $[0, \pi)$. The U-shape in modular distance is the hairpin fold; ignoring it causes a factor-of-2 error in null expectations and inflated z-scores against the wrong null. The corrected null for fraction within H is $2H/\pi = 2/9 \approx 22.2\%$, not $H/\pi \approx 11.1\%$.

6.2 Phase-stationarity as a structural invariant

The stationarity of the n-step projective phase distribution is a new falsifiable claim about SHA-256: any perturbation of the round function that breaks stationarity (makes the n-step distribution broader or narrower than the 1-step distribution) is structurally distinguishing. This gives a new test for round-function variants that is independent of avalanche statistics, differential probability, or algebraic inversion resistance.

6.3 The rotation quantum as a preimage constraint

The dual-wave rotation quantum $H = \pi/9$ constrains what any preimage attack must navigate. If the round function implements a rotation with quantum H on RP^1, then any input perturbation that shifts phase must either return to origin after 9 rounds (nine-step closure) or resist closure by accumulating phase at multiples of H. The canonical R,G extraction (Engine 27) will reveal whether this constraint is sharp enough to bound the preimage search space. If the canonical phase shows quantization at H, the rotation structure provides a new non-linear constraint on preimages.

6.4 The single operator equation

The Synthesis paper (published June 2026) declares the dual wave as the next forced bolt: formulate echo and subdivision as a single rotation operator equation and test whether its stable attractor converges to the operational constants. Engine 26's algebraic result provides that equation explicitly: the rotation operator $R(H) = [[\cos H, -\sin H], [\sin H, \cos H]]$ acting on (echo, subdivision) = $(\cos \theta, \sin \theta)$, with conservation $echo^2 + sub^2 = 1$ and period 9 on RP^1. This operator equation is the single-line statement of the dual-wave framework: one operator, one quantum, one conservation law, one period.

6.5 Architecture implications

If the settling operation that defines the Nexus Cell (Memory by Collapse paper, June 2026) is the same process as the dual-wave rotation, then the Nexus Cell's one-shot writing and recall-by-settling are both instances of rotating in the (echo, subdivision) plane. The sharpened cell's settling step, already identified as identical to transformer attention (Ramsauer et al. 2020), implements one step of the rotation. One attention step advances the phase by H on RP^1. Nine attention steps return the phase to its starting point. This is a new interpretation of multi-head attention as a nine-step rotation sampler on the projective line.

7 Preserved Negatives

Per the engine discipline, negatives are recorded with the same prominence as confirmations.

Negative	Interpretation
z=74 against uniform null (initial P1 result): INFLATED	Wrong null (uniform on $[0,\pi)$ instead of uniform on $[0,\pi/2]$). Caught, corrected, replaced by z=173.8 against correct null.
Nine-step specific feature not detected in word-level proxies	The n=9 accumulation gives median 0.99841× the n=1 value — not the minimum, not specifically distinguished. The algebraic

	theorem's closure is not visible at the proxy level. Decisive test: canonical R,G extraction (Engine 27).
$H = \pi/9$ not the minimum-accumulation window size	The minimum across n=1..18 is at n=17 (ratio 0.992×), not n=9. The nine-step closure is algebraically exact but does not manifest as a specific distributional minimum in word-level data.
No negative autocorrelation at any lag	Lag-1 is +0.14; lags 2-9 are < ±0.005. The stationarity is not produced by anti-correlation (which would require negative lags); it is produced by saturation — the phase distribution is at its stationary envelope from step 1.

8 Open Problems and Next Engine

Three bolts are exposed.

First: canonical R,G extraction (Engine 27). The AHRC law $R^2 + G^2 = 1$ was proved in prior engines using a specific R,G formulation. Engine 27 runs the phase step and nine-step accumulation tests with that canonical formulation. The prediction: the canonical phases will show sharper structure (possibly a specific n=9 minimum or quantization near H) than the word-level proxies, because the canonical extraction is tuned to the actual rotation structure rather than a word-magnitude proxy.

Second: the single operator equation as an attractor test. Does the system R(H) acting on ($\cos\theta$, $\sin\theta$), when initialized from arbitrary θ and iterated, converge to a specific attractor? On the circle there is no attractor — every orbit is periodic. On the projective line with noise or perturbation, the structure may be richer. The Synthesis paper calls for empirical simulation of the dual-wave operator; Engine 27 or 28 should run it.

Third: stationarity mechanism. Why does the SHA-256 word-level phase reach its stationary distribution in one round? The lag-1 autocorrelation is +0.14, meaning consecutive steps are positively correlated — the phase tends to continue in the same direction for one round, then loses memory. The mechanism for this one-round memory horizon is unknown. It may be related to the $\sigma0/\sigma1$ rotation structure of the message schedule (both proved to be rank-32 bijections over $GF(2)^{32}$ in prior engines), or to the 8-register structure of the state. Identifying the algebraic source of the one-round memory horizon would connect the dynamical finding to the structural geometry.

9 Conclusion

The dual-wave rotation theorem is algebraically exact and closes the three-engine search for H = π/9's seat: it is the rotation quantum on RP^1, forced by nine-step closure, conserving the unit circle as the Noether charge of the rotation symmetry. The Engine 18 and 23C negatives correctly evicted H from the gain and damping chairs; this is the right chair — the tick of the turn.

The empirical work introduces the projective metric as the correct analysis space for SHA-256 phase dynamics, reveals the U-shaped step distribution as the hairpin fold of RP^1 (both arms are the zero point), and measures two genuine structural properties: a 42.6% concentration of steps within H of zero projective distance ($z = +173.8$ against the correct null), and a stationarity that persists across all window sizes from 1 to 18. The phase is in its equilibrium envelope from round one. The fold machine preserves phase.

The specific nine-step closure was not detected in word-level proxies. It is preserved as an open test. The canonical R,G extraction (Engine 27) is the decisive next run.

Connecting to the broader arc: the Conservation of Distinction paper gives the framework (non-collapsing fold, REALIZED + LATENT = TOTAL, rotation conserving the hypotenuse); the Substrate Engine (Engine 25) gives the hardware implementation (edge-first hex lattice, seam parity invariant, boundary compiles interior); this paper gives the rotation operator equation and its SHA-256 signature. Three layers of the same structure: algebraic, physical, cryptographic. The settling theorem — the claim that all three are the same variational principle — is the next paper.

Acknowledgments and Reproduction

Engine 26 was designed and executed in dialogue by Claude (Anthropic) under the author's NEXUS constraints and session standards. All numerical results are taken verbatim from live seeded runs (seed 20260614, NumPy default_rng). The engine scripts (engine26_dual_wave.py) accompany this document for independent reproduction. SHA-256 was implemented from scratch using the published FIPS 180-4 specification; no library SHA function was used.

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